



Fig. 10: Web-based smart grid load prediction dashboard

load prediction system. The Python console loads the trained machine learning model, connects to the Firebase real-time database, retrieves transformer load data, predicts future load values, and displays time-stamped actual and predicted load readings for real-time monitoring.

Next, apply the 230V AC supply to the transformer. Once powered on, the ESP32 begins measuring the electrical parameters and uploads the data to Firebase. The transmitted data can be viewed in the Arduino IDE Serial Monitor, as shown in Fig. 7.

Fig. 7 shows the serial monitor displaying real-time electrical parameters, including voltage, current, power, energy, frequency, and power factor, measured by the PZEM-004T module. It also confirms successful Wi-Fi and Firebase connections, indicating reliable communication between the ESP32 and the cloud platform. The uploaded data serves as input to the AI-based load monitoring and prediction system, enabling real-time transformer analysis and load forecasting.

The Python program retrieves transformer data from Firebase, processes it using the trained AI model, updates the predicted values in Firebase, and stores the results in an Excel file. The corresponding output is shown in Fig. 8.

Fig. 8 shows a Microsoft Excel worksheet containing timestamped transformer parameters, including voltage, current, power, frequency, power factor, and AI-predicted load values. The automatically generated spreadsheet simplifies data visualisation, analysis, and record-keeping, while the daily, monthly, and yearly load predictions support transformer monitoring, preventive maintenance, and capacity planning.

Fig. 9 shows the Firebase real-time database storing AI-generated transformer load predictions uploaded by the Python application. The database includes predicted five-minute, daily, monthly, and yearly load values, along with peak load data, the latest update timestamp, and an overload status flag. These values are updated automatically,

enabling real-time cloud monitoring, historical analysis, and predictive maintenance.

The data can also be monitored in realtime through the web-hosted dashboard at <https://load-prediction-e14b6.web.app>, as shown in Fig. 10.

Fig. 10 shows the web-based smart grid load prediction dashboard displaying real-time transformer data and AI-generated predictions retrieved from the Firebase real-time database. The dashboard presents voltage, current, power, energy, frequency, power factor, and predicted five-minute, daily, monthly, and yearly load values, along with peak load information. It updates automatically as new data is received, providing an intuitive interface for real-time monitoring, trend analysis, overload detection, and predictive maintenance. **EFY**



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